

AUTONOMOUS FINANCIAL CRIME INVESTIGATION AGENT

FinCrime AI — SIH Presentation Knowledge Base

Deep Research Report · All 6 Slides Covered

IBM HI-Small AML Dataset · LangGraph · Gemini 2.5 Flash · FastAPI · React

SLIDE 1 · TITLE SLIDE

Project Identity

- **Project Title:** Autonomous Financial Crime Investigation Agent
- **Short Name:** FinCrime AI
- **Theme:** Fintech / AI-powered Financial Security
- **PS Category:** SIH Problem Statement PS-2 — AML / Financial Crime Investigation
- **Dataset:** IBM HI-Small AML Dataset — 370 verified ground-truth laundering patterns

Problem Statement

Banks and Financial Intelligence Units face an overwhelming volume of suspicious transaction alerts. Manual investigation is slow, inconsistent, and unscalable. Investigators must query multiple systems (transaction history, KYC records, complaints, linked accounts) to build an evidence picture — a process that takes hours per case. The system automates this investigation pipeline using an AI agent while keeping the final decision with a human analyst.

SLIDE 2 · IDEA / PROPOSED SOLUTION

What We Are Building

- A full-stack AI system that ingests financial transactions in real-time
- Uses ML models to detect suspicious transactions automatically
- Triggers an AI investigation agent (LangGraph state machine) that gathers evidence from 4+ data sources
- Detects money laundering typologies algorithmically: FAN-OUT, FAN-IN, CYCLE, STACK, BIPARTITE, SCATTER-GATHER
- Generates a structured, evidence-backed investigation report with risk score (0–100), risk level, typology, and recommendation
- Presents everything on a React analyst dashboard where the human makes the final decision

How It Solves The Problem

- Replaces hours of manual investigation with an automated agent completing evidence collection in seconds
- Agent never hallucinates — uses only observable, collected evidence from real data sources
- AI recommendation is advisory; human analyst always decides (ESCALATE / CLEAR / FALSE POSITIVE / REVIEW)
- Full audit trail of every agent action, tool call, and human decision is logged to database

What Makes It Innovative

Innovation	Description
LangGraph Agentic State Machine	Not a simple chatbot. Agent dynamically selects which tools to call based on evidence already collected.
Two-Stage ML Pipeline	Fast baseline model (5 features) screens all transactions; enhanced model (14 features) runs only on borderline cases.
Deterministic Pattern Detection	FAN-OUT, FAN-IN, CYCLE, STACK, BIPARTITE detected algorithmically — fully auditable and explainable.
Evidence Correlation Engine	Transparent risk breakdown — every point is itemized (KYC incomplete=+20, CYCLE pattern=+45, etc.)
Gemini 2.5 Flash (Advisory Only)	Used only for investigation planning and narrative explanation, never for risk scoring.
SCN-005 False Positive	System correctly clears legitimate high-value transactions — demonstrates FP handling.

SLIDE 3 · TECHNICAL APPROACH

Tech Stack

Layer	Technology
Frontend Framework	React 19 + Vite 8
Frontend Styling	Tailwind CSS 3
Frontend Libraries	Framer Motion, ReactFlow, Recharts, Lucide-React, React Router v7
Backend	Python + FastAPI 0.115 + Uvicorn
Database	MySQL (InnoDB) — 7 tables
ML Models	scikit-learn (Random Forest), joblib, pandas, numpy
Agent Orchestration	LangGraph 0.2
LLM	Google Gemini 2.5 Flash (google-genai SDK)
Data Validation	Pydantic v2
Deployment	Vercel (frontend) + Uvicorn (backend)

8-Layer Architecture

■ Layer 1 — Data

IBM HI-Small dataset (370 AML patterns) + synthetic KYC, complaints, account relationships CSVs

■ Layer 2 — Detection

Two-stage ML: Baseline model (5 features) → if 20–64, run Enhanced model (14 features including KYC, complaints, network). Thresholds: baseline $\geq$ 65 or enhanced $\geq$ 45 creates a case.

■ Layer 3 — Investigation Agent

LangGraph StateGraph: START → agent_node → (tool nodes loop) → pattern_analysis_node → risk_assessment_node → END

■ Layer 4 — Investigation Tools

get_transactions, get_kyc, get_linked_accounts, get_complaints, get_transaction_graph, get_scenario_transactions — data from CSV or live MySQL API

■ Layer 5 — Pattern Detection

Deterministic algorithms: FAN-OUT (min 3 destinations), FAN-IN (min 3 sources), CYCLE (DFS up to 10 hops), STACK (3-node chains), SCATTER-GATHER, BIPARTITE (disjoint groups)

■ Layer 6 — Risk Scoring

Evidence correlation engine — every signal has a documented point weight; MAX 100; LOW $<$ 40, MEDIUM 40-74, HIGH $\geq$ 75

■ Layer 7 — FastAPI REST API

Routes: /cases, /transactions, /accounts/{id}/*, /ml/score, /audit — auto-triggers investigation as background task when ML flags suspicious transaction

■ Layer 8 — React Dashboard

Pages: Dashboard, AML Demo, Live Investigation (real-time), Case Detail, Audit Trail

LangGraph Agent Flow (Step by Step)

- START → agent_node: Reads case state, checks scenario ID
- Controlled scenario: loads ground-truth evidence via get_scenario_transactions
- Demo mode (demo_mode=True): uses deterministic 4-tool plan
- Live mode: calls Gemini 2.5 Flash create_investigation_plan tool-call to get ordered tool list
- Routes to tool node: transactions / kyc / linked_accounts / complaints / counterparty_kyc / counterparty_complaints
- Each tool node appends result to state['observations'] and marks tool as used in state['used_tools']
- Returns to agent_node; picks next tool from plan until all used
- All tools used → routes to pattern_analysis_node (runs all 6 detectors)
- risk_assessment_node: calls correlate_evidence(observations) → final risk score, level, typology, recommendation
- Builds final report dict → stored in state['report'] → END

Risk Scoring Breakdown

Risk Factor	Points
Multiple unique receivers	+10
Multiple unique senders	+10
Incomplete KYC	+20
High KYC risk rating	+20
New account (< 30 days)	+20
Linked accounts (5 per, max 20)	+5–20
Complaints (20 per, max 20)	+20
FAN-OUT pattern	+40
FAN-IN pattern	+40
CYCLE pattern	+45
STACK pattern	+40
BIPARTITE pattern	+40
MAX SCORE	100 (capped)

Risk Level: LOW (score < 40) → CLEAR | MEDIUM (40–74) → REVIEW | HIGH (≥ 75) → ESCALATE

SLIDE 4 · FEASIBILITY & VIABILITY

Why It Can Actually Be Built

- Already largely built — LangGraph agent, ML models, FastAPI backend, React frontend all implemented and functional
- Real dataset — IBM HI-Small AML dataset with 370 verified ground-truth patterns; publicly available
- No LLM hallucination risk — pattern detection and risk scoring are deterministic Python algorithms
- Modular architecture — each layer independently replaceable (swap sklearn for XGBoost, add Neo4j graph DB)
- Demo-ready — 5 controlled scenarios cover FAN-OUT, CYCLE, STACK, BIPARTITE, FALSE POSITIVE end-to-end
- Graceful degradation — all external dependencies (Gemini, MySQL) have fallbacks (CSV data, deterministic plans, local persistence)
- Auto-retry — agent result submission retried 3× via API before local repository fallback

Challenges & Mitigations

Challenge	Mitigation Strategy
LLM planning may select wrong tools	Deterministic fallback plan always available; Gemini is advisory, never required
Database connectivity during demo	CSVs serve as full offline data source; backend API toggled at runtime
ML model accuracy on unseen data	Two-stage pipeline with conservative thresholds; human analyst reviews before any final action
Privacy of real transaction data	Only IBM HI-Small (public) + fully synthetic KYC/complaints used
Scalability of graph traversal	Currently NetworkX in-memory; can scale to Neo4j for production
False positives overwhelming analysts	SCN-005 demonstrates correct FP clearing; ML threshold tuned (65/45/20)
Audit / regulatory compliance	Every action logged to audit_logs with actor, timestamp, and details

SLIDE 5 · IMPACT & BENEFITS

Who Benefits

Beneficiary	How They Benefit
Bank AML / Compliance Teams	Automates the most time-consuming part of investigation — evidence gathering across 4+ sources
Financial Intelligence Units (FIUs)	Receive structured, evidence-backed reports for escalated cases instead of raw alerts
Human Analysts	Freed from repetitive data collection; focus on judgment and final decision-making
End Customers	Legitimate high-value transactions (like SCN-005 — ₹71.47 Cr cheque) get cleared faster and correctly
Regulators / Auditors	Full audit trail of every agent action and human decision with timestamps

Measurable Impact

- **Speed:** Agent completes evidence collection across 4+ sources in seconds vs hours manually
- **Consistency:** Every investigation follows the same evidence-gathering protocol — no investigator bias
- **Explainability:** Every risk point is itemized in the risk breakdown — auditors can trace every score contribution
- **Auditability:** Every action (tool call, agent decision, human decision) logged to audit_logs table
- **False Positive Reduction:** Two-stage ML pipeline + pattern-based verification reduces unnecessary escalations

Social / Economic / Environmental Benefits

- **Social:** Reduces financial crime that funds terrorism, trafficking, and organized crime
- **Economic:** AML compliance costs globally ~\$270 billion/year — automation can cut this significantly for banks
- **Regulatory:** Supports compliance with PMLA (Prevention of Money Laundering Act) in India and FATF guidelines
- **Environmental:** Digital investigation replaces paper-based, manual processes

SLIDE 6 · RESEARCH & REFERENCES

Datasets

- **IBM HI-Small AML Dataset:** 370 verified money laundering patterns across FAN-OUT, FAN-IN, CYCLE, STACK, SCATTER-GATHER, BIPARTITE typologies. Source: IBM Research / Kaggle
- **Synthetic Supporting Data:** KYC records, complaint logs, account relationships — generated to support 5 demo scenarios
- **Total ground-truth patterns used in demo:** 4 patterns (SCN-001 to SCN-004) + 1 false-positive scenario (SCN-005)

APIs & Libraries Used

Library / API	Version	Purpose
LangGraph	0.2	Agentic state machine — investigation orchestration
Google Gemini 2.5 Flash	google-genai 1.x	Investigation planning + narrative explanation (advisory)
FastAPI	0.115	REST API backend
scikit-learn	1.5	Random Forest — baseline and enhanced fraud models
pandas / numpy	2.2 / 1.26	Feature engineering and data processing
joblib	1.4	ML model serialization and loading
React	19	Frontend analyst dashboard
ReactFlow	11.11	Network graph visualization
Recharts	3.10	Transaction charts
Framer Motion	13.1	Animations and transitions
Pydantic v2	2.10	Data validation and API contracts
MySQL (InnoDB)	—	Persistent storage — 7 tables

Technical References

- FATF (Financial Action Task Force) AML Typology Reports — FAN-OUT, CYCLE, STACK, BIPARTITE typologies align with FATF definitions
- Two-stage ML screening methodology follows best practices for high-recall fraud detection (flag more, then filter)
- Evidence-based risk scoring with itemized breakdown aligns with FinCEN (Financial Crimes Enforcement Network) investigation guidance
- LangGraph framework: <https://langchain-ai.github.io/langgraph/>
- Google Gemini API: <https://ai.google.dev/>

Government Sources

- **PMLA (Prevention of Money Laundering Act, 2002)** — Indian legal framework for AML investigations
- **FATF Recommendations** — International standards for anti-money laundering and counter-terrorism financing

- **RBI AML/CFT Circulars** — Reserve Bank of India guidelines for Anti-Money Laundering and Combating the Financing of Terrorism
- **FIU-IND** — Financial Intelligence Unit India, under Ministry of Finance

5 Demo Scenarios (Ground-Truth Backed)

Scenario	Name	Typology	Key Detail
SCN-001	Fan-Out Investigation	FAN-OUT	Account 021174:800737690 sends to 16 unique destinations
SCN-002	Cycle Investigation	CYCLE	Account 01467:8013C4030 — up to 10-hop transaction cycle
SCN-003	Stack Investigation	STACK	Multiple independent 3-node transaction chains (A→B→C)
SCN-004	Bipartite Network	BIPARTITE	Disjoint sender/receiver groups with 4+ cross-edges
SCN-005	False Positive	NONE	0112733:804BD7DC0 — ■71.47 Cr cheque correctly CLEARED